

Chapter 5

Network Layer:

The Control Plane

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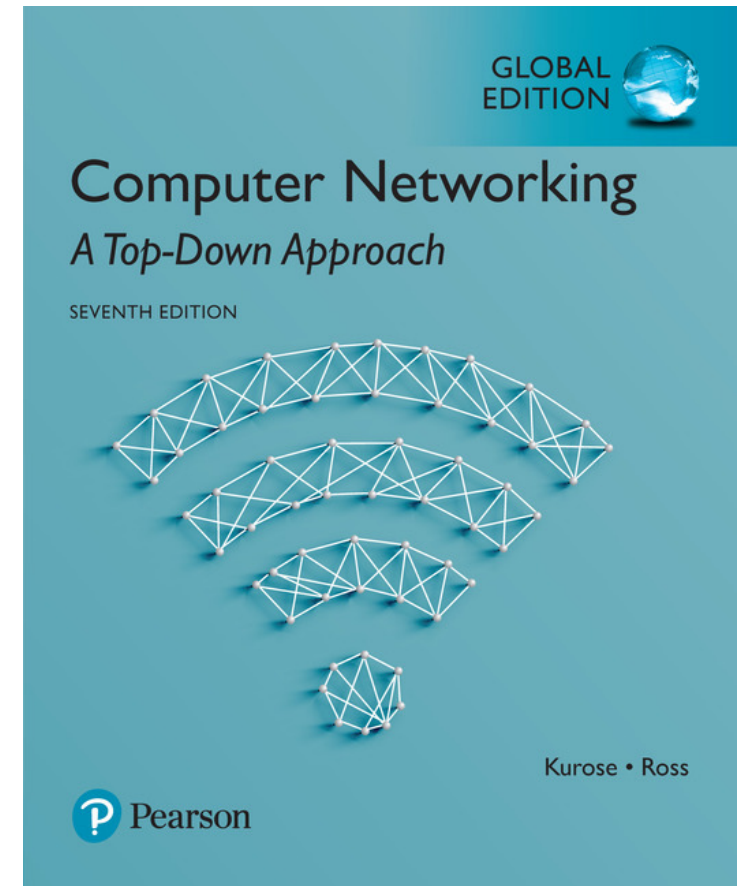
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Computer Networking: A Top Down Approach

7th Edition, Global Edition
Jim Kurose, Keith Ross
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Chapter 5: network layer control plane

chapter goals: understand principles behind network control plane

- traditional routing algorithms
- SDN controllers
- Internet Control Message Protocol
- network management

and their instantiation, implementation in the Internet:

- OSPF, BGP, OpenFlow, ODL and ONOS controllers, ICMP, SNMP

Chapter 5: outline

5.1 introduction

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet: OSPF

5.4 routing among the ISPs: BGP

5.5 The SDN control plane

5.6 ICMP: The Internet Control Message Protocol

5.7 Network management and SNMP

Network-layer functions

Recall: two network-layer functions:

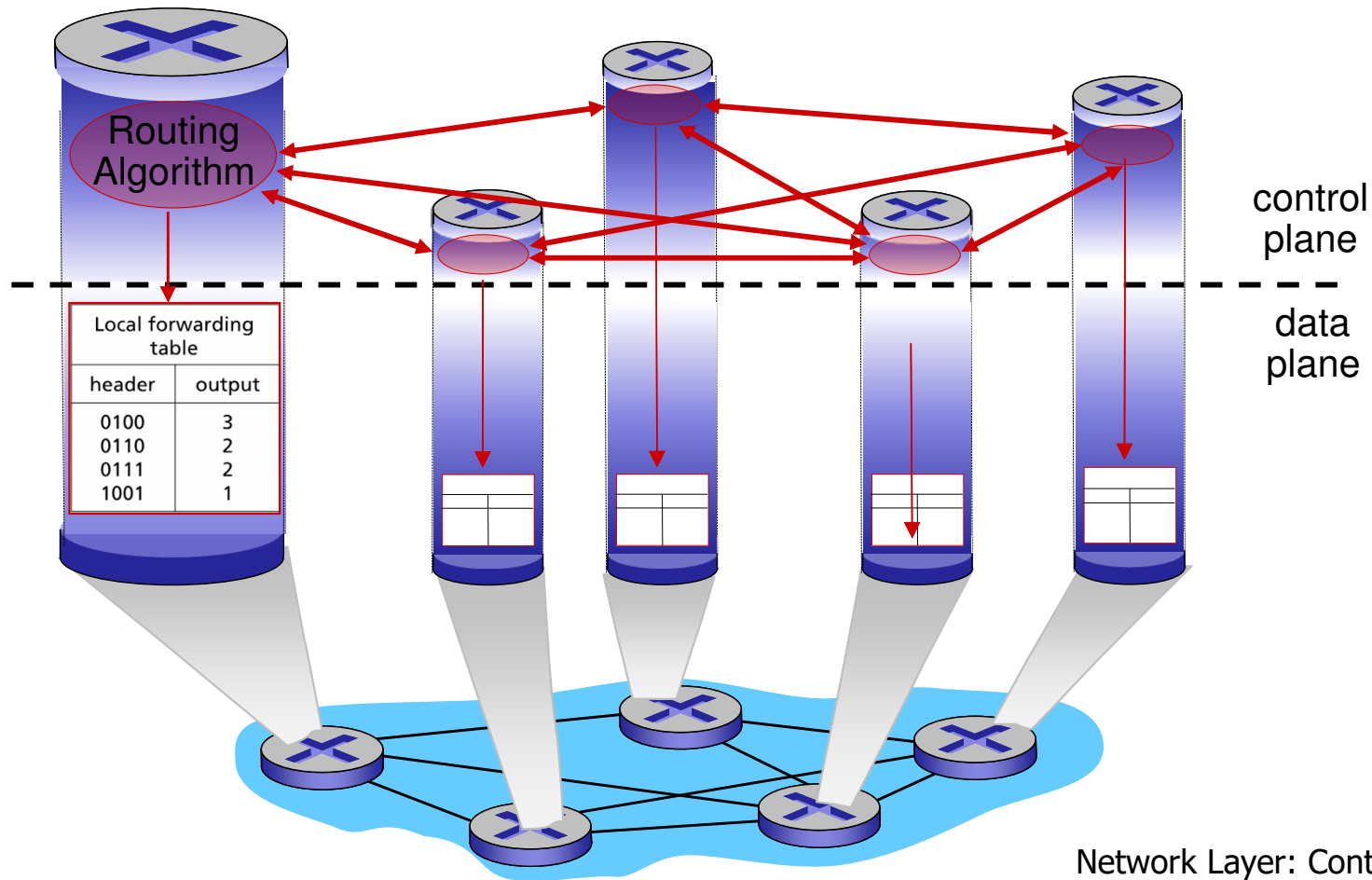
- *forwarding*: move packets from router's input to appropriate router output *data plane*
- *routing*: determine route taken by packets from source to destination *control plane*

Two approaches to structuring network control plane:

- per-router control (traditional)
- logically centralized control (software defined networking)

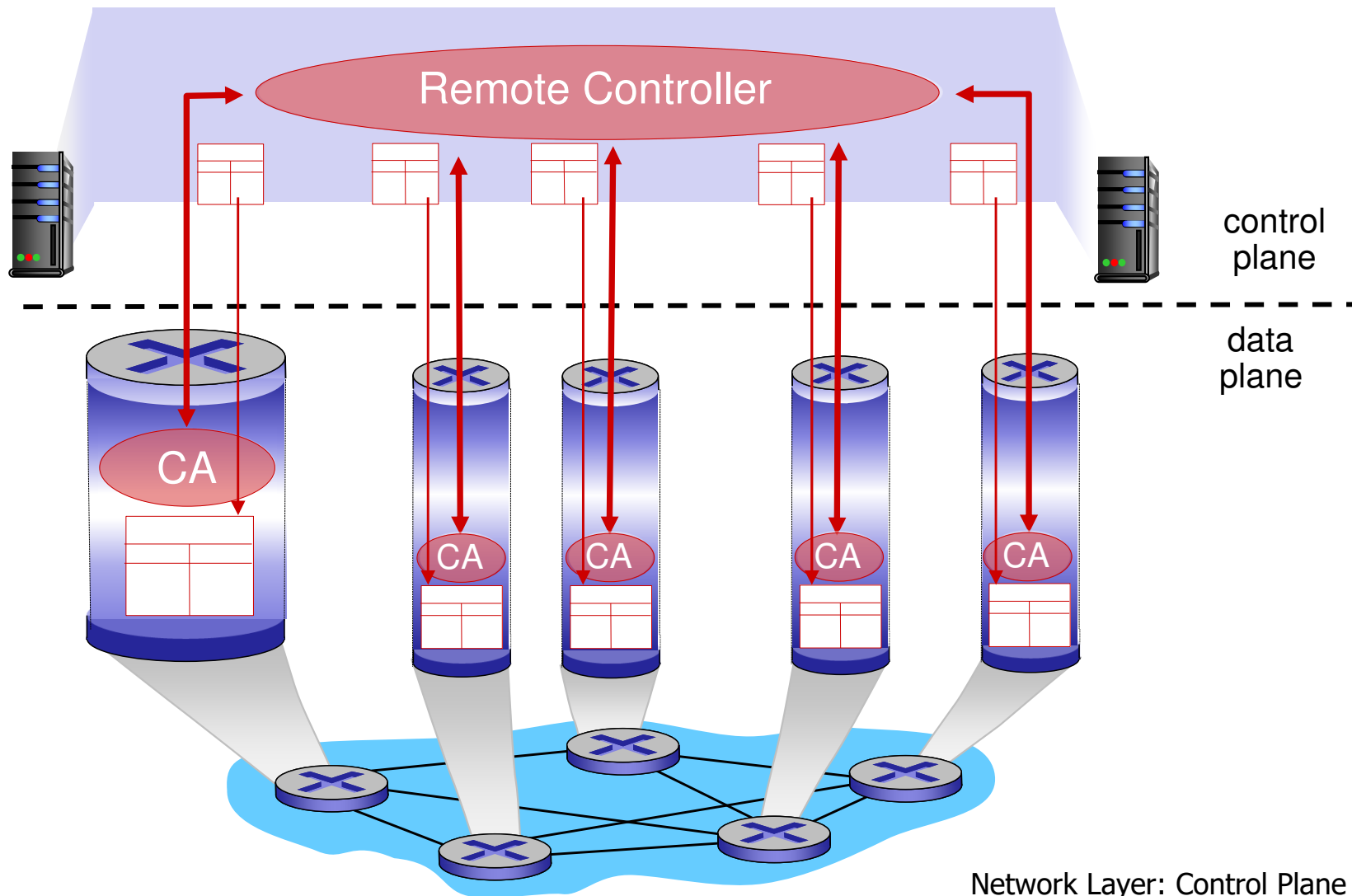
Per-router control plane

Individual routing algorithm components *in each and every router* interact with each other in control plane to compute forwarding tables



Logically centralized control plane

A distinct (typically remote) controller interacts with local control agents (CAs) in routers to compute forwarding tables



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5.3 intra-AS routing in the Internet: OSPF

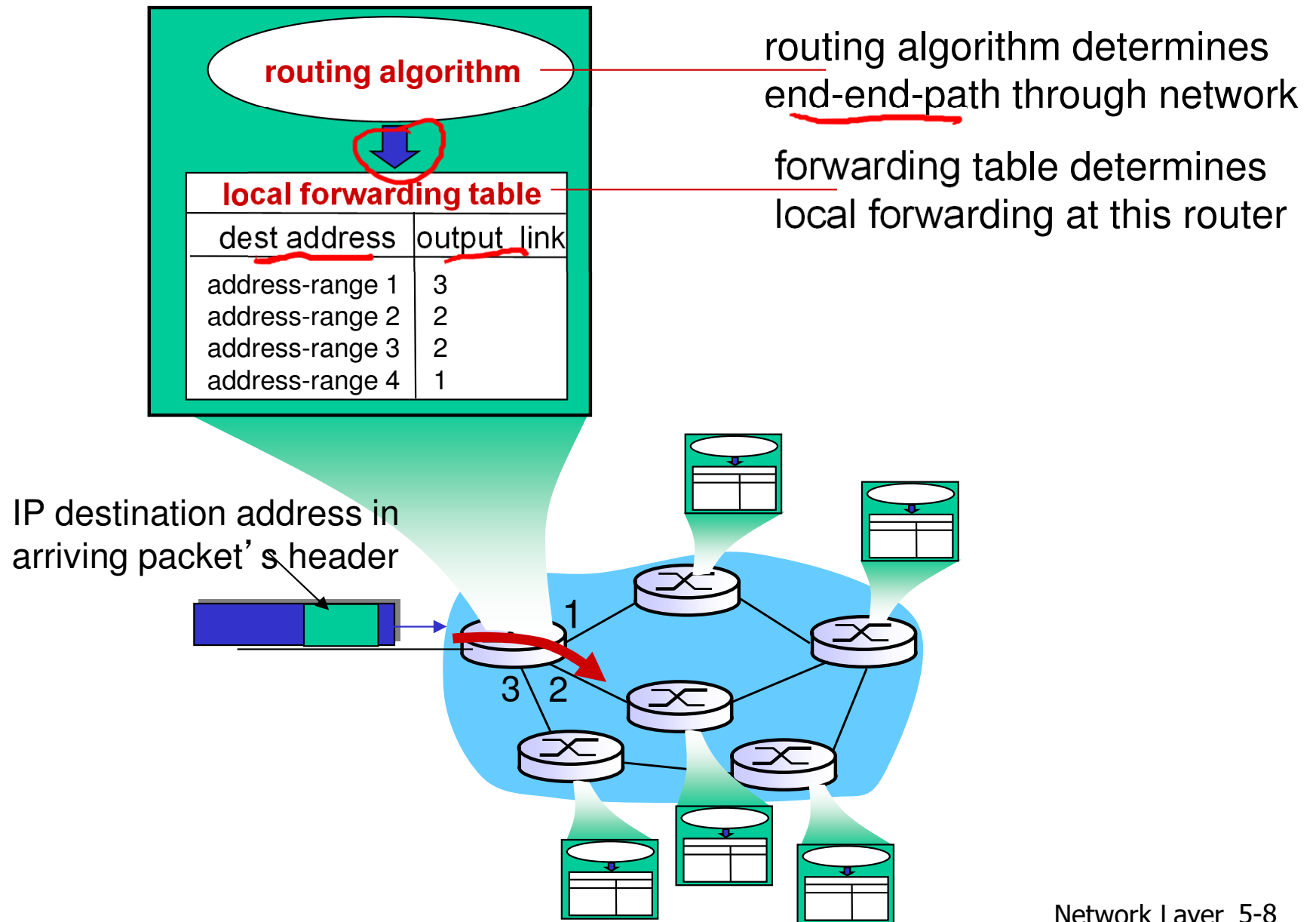
5.4 routing among the ISPs: BGP

5.5 The SDN control plane

5.6 ICMP: The Internet Control Message Protocol

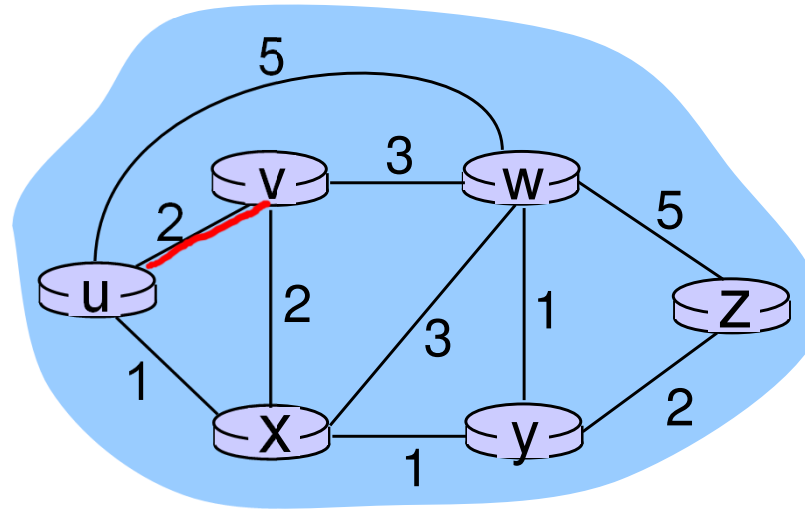
5.7 Network management and SNMP

Interplay between routing, forwarding



Graph abstraction

Graph theory



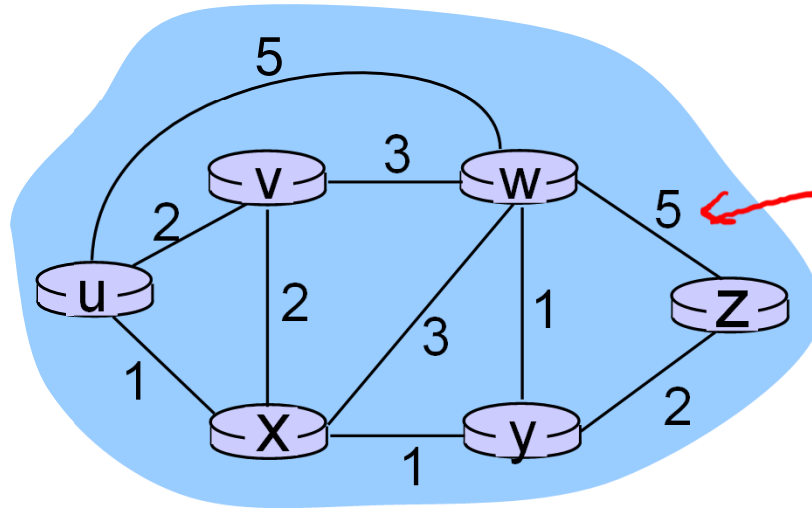
graph: $G = (N, E)$

$N = \text{set of routers} = \{ u, v, w, x, y, z \}$

$E = \text{set of links} = \{ (u,v), (u,x), (v,x), (v,w), (x,w), (x,y), (w,y), (w,z), (y,z), (u,w) \}$

aside: graph abstraction is useful in other network contexts, e.g., P2P, where N is set of peers and E is set of TCP connections

Graph abstraction: costs



$c(x, x')$ = cost of link (x, x')
e.g., $c(w, z) = 5$

cost could always be 1, or
inversely related to bandwidth,
or inversely related to
congestion

$$\text{cost of path } (x_1, x_2, x_3, \dots, x_p) = c(x_1, x_2) + c(x_2, x_3) + \dots + c(x_{p-1}, x_p)$$

key question: what is the least-cost path between u and z?

Solution: Routing algorithm that finds that least cost path

Routing algorithm classification

Q: Centralized or decentralized?

centralized:

ans ❖ all routers have complete topology, link cost info

algo ❖ “link state” algorithms *Dijkstra*

decentralized:

ans ❖ router knows physically-connected neighbors, link costs to neighbors

❖ iterative process of computation, exchange of info with neighbors

algo ❖ “distance vector” algorithms *Bellman-Ford*

Q: static or dynamic?

static:

❖ routes change slowly over time

dynamic:

❖ routes change more quickly

- periodic update
- in response to link cost changes

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A Link-State Routing Algorithm

Dijkstra's algorithm

- ❖ network topology & link costs known to all nodes
 - Complete Info
 - accomplished via “link state broadcast”
 - all nodes have same info
- ❖ computes least cost paths from one node (‘source’) to all other nodes
 - by-product
 - gives *forwarding table* for that node
- ❖ iterative: after k iterations, know least cost path to k dest.’s

notation:

- ❖ $C(x,y)$: link cost from node x to y; $= \infty$ if not direct neighbors
- ❖ $D(v)$: current value of cost of path from source to dest. v
- ❖ $p(v)$: predecessor node along path from source to v
- ❖ N' : set of nodes whose least cost path definitively known

Dijkstra's Algorithm

Source node is u

1 **Initialization:**

2 $N' = \{u\}$

3 for all nodes v

4 if v adjacent to u

5 then $D(v) = c(u,v)$

6 else $D(v) = \infty$

7

8 **Loop**

9 ✓ find w not in N' such that $D(w)$ is a minimum

10 ✓ add w to N'

11 ✓ update $D(v)$ for all v adjacent to w and not in N' :

12 ✓ **$D(v) = \min(D(v), D(w) + c(w,v))$**

13 /* new cost to v is either old cost to v or known

14 shortest path cost to w plus cost from w to v */

15 **until all nodes in N'**

Dijkstra's algorithm: example

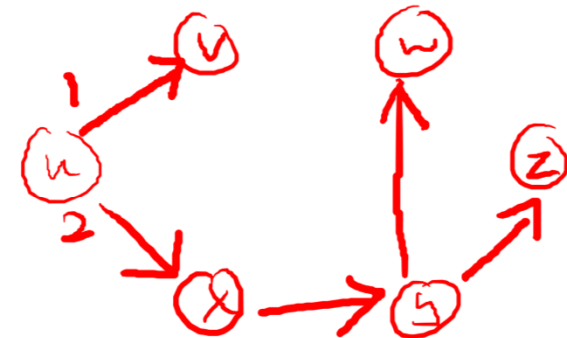
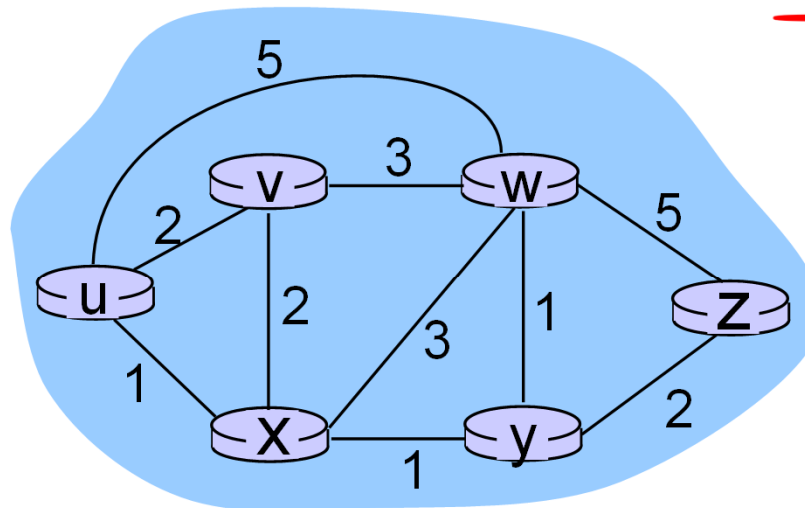
Route

Step	N'	^v D(v),p(v)	^w D(w),p(w)	^x D(x),p(x)	^y D(y),p(y)	^z D(z),p(z)
0	u	2,u	5,u	<u>1,u</u>	∞	∞
1	ux ←	2,u	4,x		<u>2,x</u>	∞
2	uxy ←	<u>2,u</u>	3,y			4,y
3	uxyv ←		<u>3,y</u>			4,y
4	uxyvw ←					<u>4,y</u>
5	uxyvwz ←					

Shortest-path tree

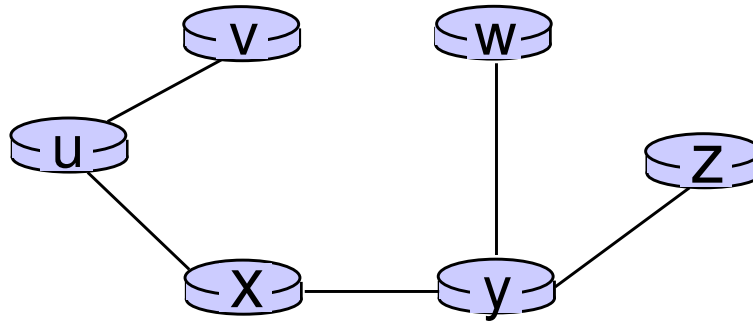
Forwarding table

Dest	Interface
v	1
x	2
y	2
w	2
z	2



Dijkstra's algorithm: example

resulting shortest-path tree from u:



resulting forwarding table in u:

destination	link
v	(u,v)
x	(u,x)
y	(u,x)
w	(u,x)
z	(u,x)

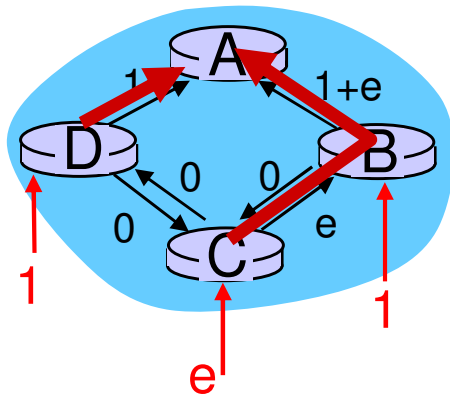
Dijkstra's algorithm, discussion

algorithm complexity: n nodes

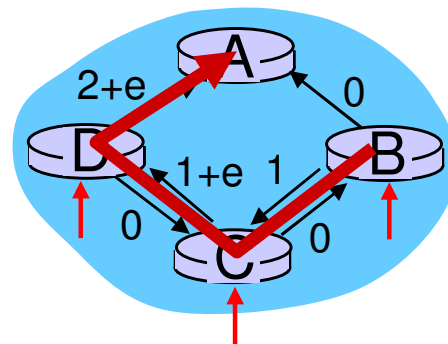
- ❖ each iteration: need to check all nodes, w, not in N
- ❖ $n(n+1)/2$ comparisons: $O(n^2)$
- ❖ more efficient implementations possible: $O(n \log n)$

oscillations possible:

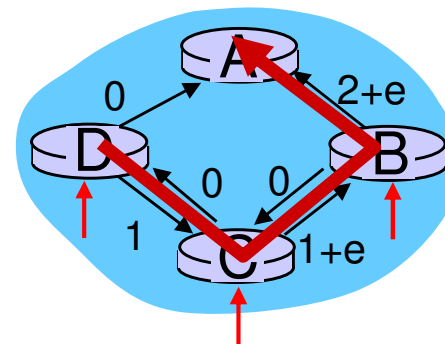
- ❖ e.g., Congestion-sensitive routing: suppose link cost equals amount of carried traffic



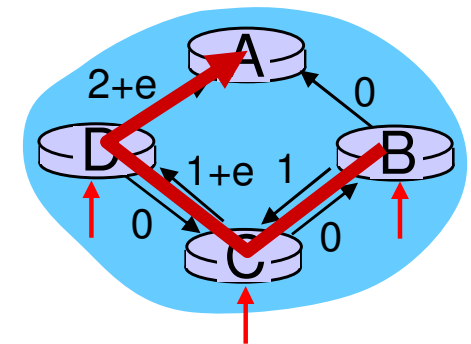
initially



given these costs,
find new routing....
resulting in new costs



given these costs,
find new routing....
resulting in new costs



given these costs,
find new routing....
resulting in new costs

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Distance vector algorithm

Bellman-Ford equation (dynamic programming)

let

$d_x(y) :=$ cost of least-cost path from x to y

then

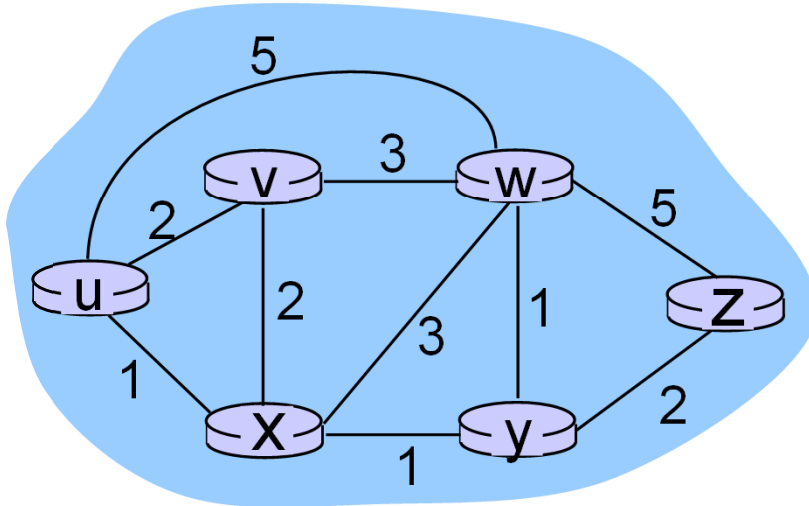
$$d_x(y) = \min_v \{ \underbrace{c(x,v)}_{\text{cost to neighbor } v} + \underbrace{d_v(y)}_{\text{cost from neighbor } v \text{ to destination } y} \}$$

$x \xrightarrow{\text{neighbour}} v \xrightarrow{\quad} y$

min taken over all neighbors v of x

Bellman-Ford example

Dijkstra's algo perspective



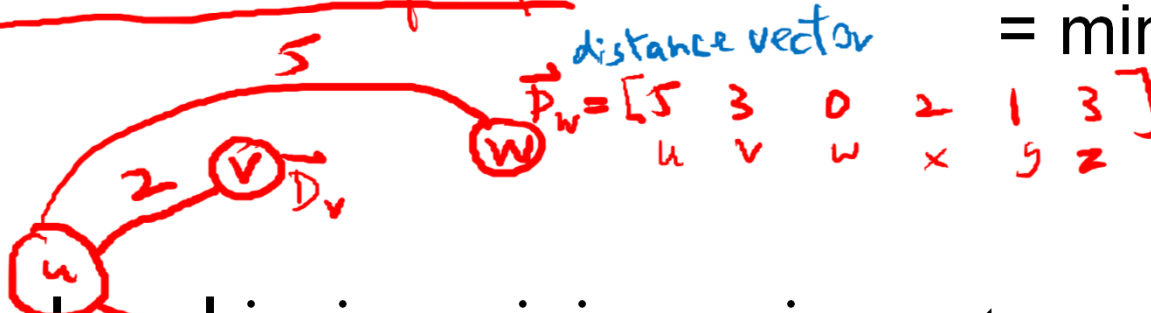
Given $d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

B-F equation says:

$$d_u(z) = \min \{ c(u, v) + d_v(z), c(u, x) + d_x(z), c(u, w) + d_w(z) \}$$

$$= \min \{ 2 + 5, 1 + 3, 5 + 3 \} = \underline{4}$$

Bellman-Ford's perspective



node achieving minimum is next hop in shortest path, used in forwarding table

Distance vector algorithm

- ❖ $D_x(y)$ = estimate of least cost from x to y
 - x maintains distance vector $D_x = [D_x(y): y \in N]$
- ❖ node x:
 - knows cost to each neighbor v: $c(x,v)$
 - maintains its neighbors' distance vectors. For each neighbor v, x maintains $D_v = [D_v(y): y \in N]$

Distance vector algorithm

key idea:

- ❖ from time-to-time, each node sends its own distance vector estimate to neighbors
- ❖ when x receives new DV estimate from neighbor, it updates its own DV using B-F equation:

$$D_x(y) \leftarrow \min_v \{c(x, v) + D_v(y)\} \text{ for each node } y \in N$$

- ❖ under minor, natural conditions, the estimate $D_x(y)$ converge to the actual least cost $d_x(y)$

Distance vector algorithm

iterative, asynchronous:

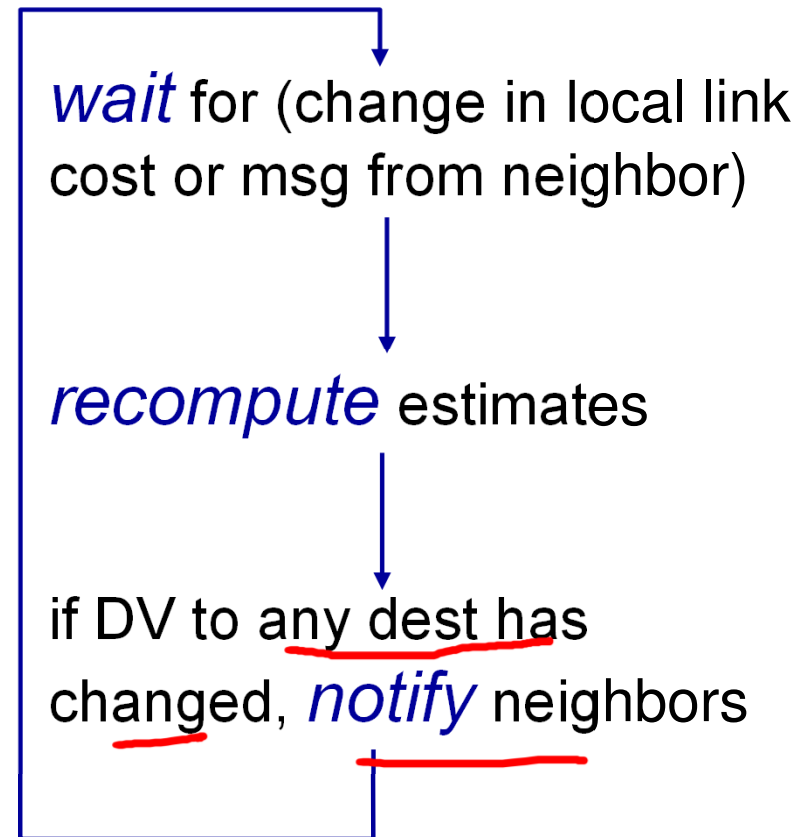
each local iteration
caused by:

- ❖ local link cost change
- ❖ DV update message from neighbor

distributed:

- ❖ each node notifies neighbors *only* when its DV changes
 - neighbors then notify their neighbors if necessary

each node:



$$D_x(y) = \min\{c(x,y) + D_y(y), c(x,z) + D_z(y)\}$$

$$= \min\{2+0, 7+1\} = 2$$

$$D_x(z) = \min\{c(x,y) + D_y(z), c(x,z) + D_z(z)\}$$

$$= \min\{2+1, 7+0\} = 3$$

$D_x(z)$
node x
table

		cost to		
		x	y	z
from	x	0	2	7
	y	∞	∞	∞
	z	∞	∞	∞

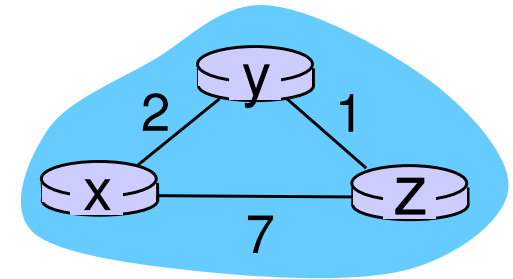
		cost to		
		x	y	z
from	x	0	2	3
	y	2	0	1
	z	7	1	0

node y
table

		cost to		
		x	y	z
from	x	∞	∞	∞
	y	2	0	1
	z	∞	∞	∞

node z
table

		cost to		
		x	y	z
from	x	∞	∞	∞
	y	∞	∞	∞
	z	7	1	0



time

$$D_x(y) = \min\{c(x,y) + D_y(y), c(x,z) + D_z(y)\}$$

$$= \min\{2+0, 7+1\} = 2$$

$$D_x(z) = \min\{c(x,y) + D_y(z), c(x,z) + D_z(z)\}$$

$$= \min\{2+1, 7+0\} = 3$$

node x
table

		cost to		
		x	y	z
from	x	0	2	7
	y	∞	∞	∞
	z	∞	∞	∞

node y
table

		cost to		
		x	y	z
from	x	∞	∞	∞
	y	2	0	1
	z	∞	∞	∞

node z
table

		cost to		
		x	y	z
from	x	∞	∞	∞
	y	∞	∞	∞
	z	7	1	0

		cost to		
		x	y	z
from	x	0	2	3
	y	2	0	1
	z	7	1	0

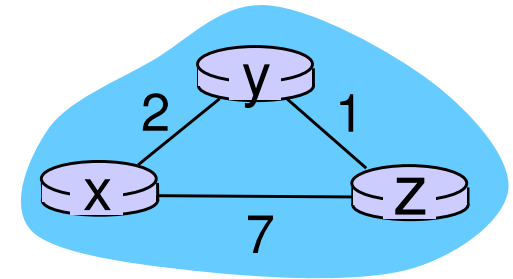
		cost to		
		x	y	z
from	x	0	2	7
	y	2	0	1
	z	7	1	0

		cost to		
		x	y	z
from	x	0	2	7
	y	2	0	1
	z	3	1	0

		cost to		
		x	y	z
from	x	0	2	3
	y	2	0	1
	z	3	1	0

		cost to		
		x	y	z
from	x	0	2	3
	y	2	0	1
	z	3	1	0

		cost to		
		x	y	z
from	x	0	2	3
	y	2	0	1
	z	3	1	0

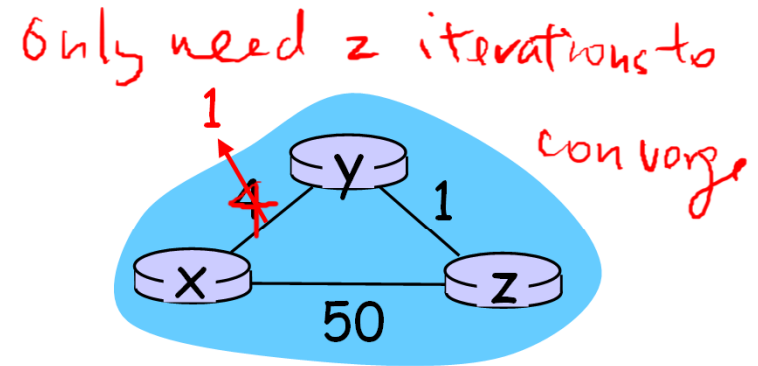


time

Distance vector: link cost changes

link cost changes:

- ❖ node detects local link cost change
- ❖ updates routing info, recalculates distance vector
- ❖ if DV changes, notify neighbors



“good
news
travels
fast”

t_0 : y detects link-cost change, updates its DV, informs its neighbors.

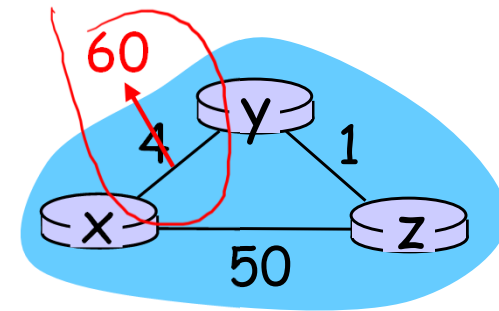
t_1 : z receives update from y, updates its table, computes new least cost to x, sends its neighbors its DV.

t_2 : y receives z's update, updates its distance table. y's least costs do *not* change, so y does *not* send a message to z.

Distance vector: link cost changes

link cost changes:

- ❖ node detects local link cost change
- ❖ *bad news travels slow* - “count to infinity” problem!
- ❖ 44 iterations before algorithm stabilizes: see text



Application ^{OSPF} ^{RIP} Comparison of LS and DV algorithms

1) message complexity

- ❖ **LS:** with n nodes, E links, $O(nE)$ msgs sent *complete topology info*
- ❖ **DV:** exchange between neighbors only
 - convergence time varies

2) speed of convergence

- ❖ **LS:** $O(n^2)$ algorithm requires $O(nE)$ msgs
 - may have oscillations
- ❖ **DV:** convergence time varies
 - may be routing loops
 - count-to-infinity problem

3) robustness: what happens if router malfunctions?

LS:

- node can advertise incorrect link cost *affect itself*
- each node computes only its own table

DV:

- DV node can advertise incorrect path cost
- each node's table used by others
 - error propagate thru network

affect others

✓ but speed varies

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Hierarchical routing

our routing study thus far - idealization

Theory

- ❖ all routers identical
- ❖ network “flat”

... not true in practice

Practice

scale: with 600 million destinations:

- can't store all dest's in routing tables!
- routing table exchange would swamp links!

administrative autonomy

- ❖ internet = network of networks
- ❖ each network admin may want to control routing in its own network

Internet approach to scalable routing

aggregate routers into regions known as “autonomous systems” (AS) (a.k.a. “domains”)

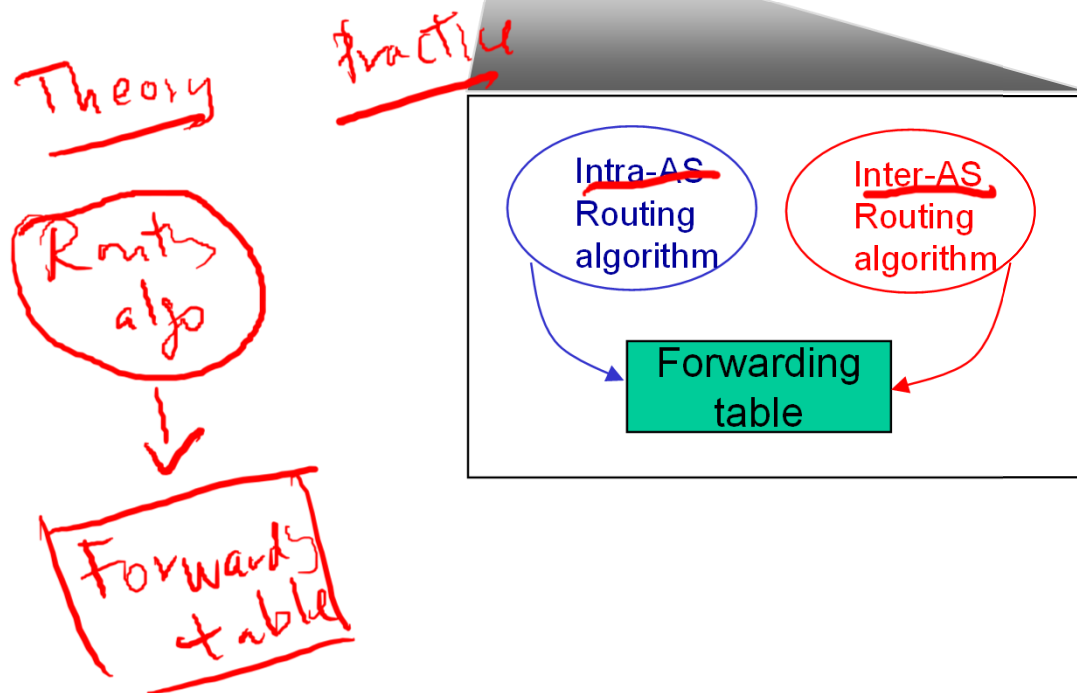
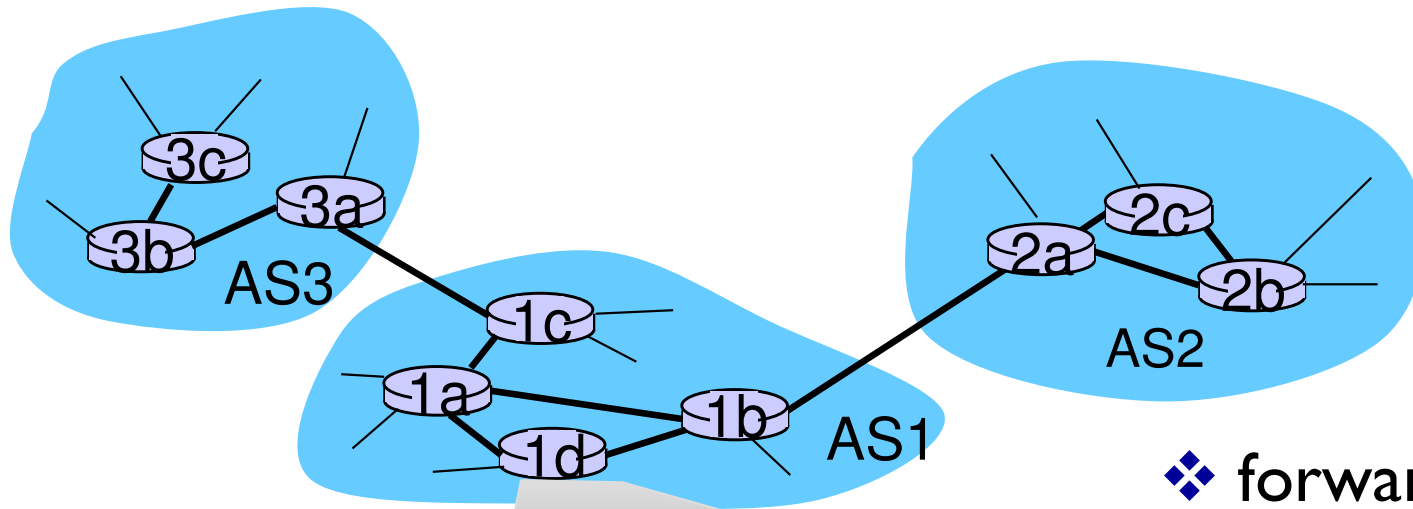
intra-AS routing

- routing among hosts, routers in same AS (“network”)
- all routers in AS must run same intra-domain protocol
- routers in *different AS* can run *different* intra-domain routing protocol
- gateway router: at “edge” of its own AS, has link(s) to router(s) in other AS'es

inter-AS routing

- routing among AS'es
- gateways perform inter-domain routing (as well as intra-domain routing)

Interconnected ASes



❖ forwarding table configured by both intra- and inter-AS routing algorithm

- **intra-AS** sets entries for **internal** destinations
- **inter-AS & intra-AS** sets entries for **external** destinations

Intra-AS Routing

- ❖ also known as *interior gateway protocols (IGP)*
- ❖ most common intra-AS routing protocols:
 - RIP: Routing Information Protocol
 - OSPF: Open Shortest Path First
 - IGRP: Interior Gateway Routing Protocol (Cisco proprietary)

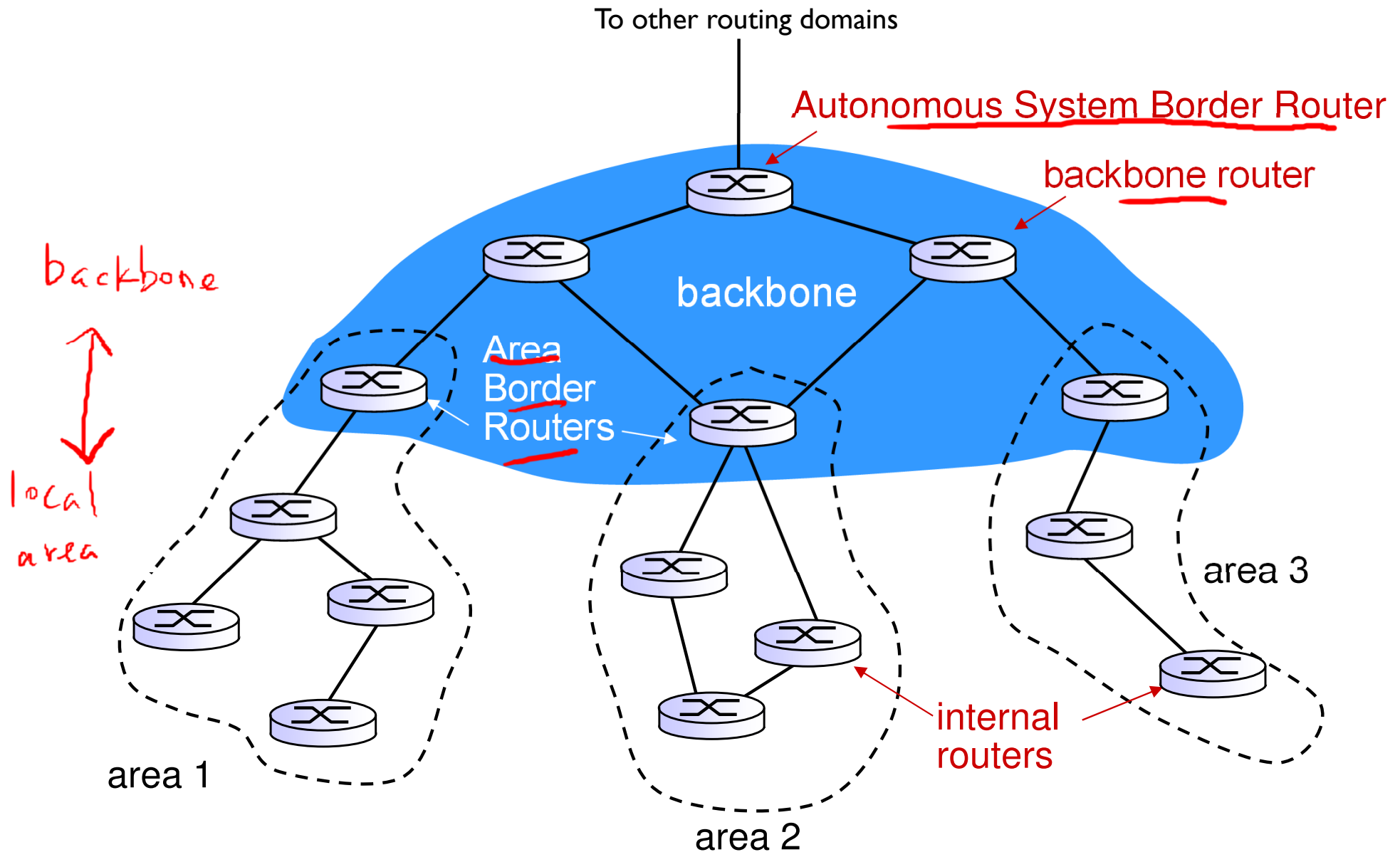
OSPF (Open Shortest Path First)

- “open”: publicly available
- uses link state algorithm:
 1. Floods link state information
 2. Constructs a complete topology map of the entire AS
 3. Runs Dijkstra's algorithm to determine a shortest-path tree to all subnets, with itself as the root node
- advertisements flooded to *entire* AS
 - carried in OSPF messages directly over IP (rather than TCP or UDP)

Hierarchical OSPF

- two-level hierarchy:
 - multiple local areas, one backbone.
 - link-state advertisements only within an area
 - each nodes has detailed area topology; only know direction (shortest path) to networks in other areas.
- *Area Border Routers (ABR):*
 - “summarize” distances to networks in own area, advertise to other Area Border routers.
- *Autonomous System Border Router (ASBR):*
 - connect to other routing domain (e.g., RIP, BGP) and/or AS.
- *Backbone routers:*
 - Run OSPF routing limited to backbone.

Hierarchical OSPF



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Inter-AS tasks

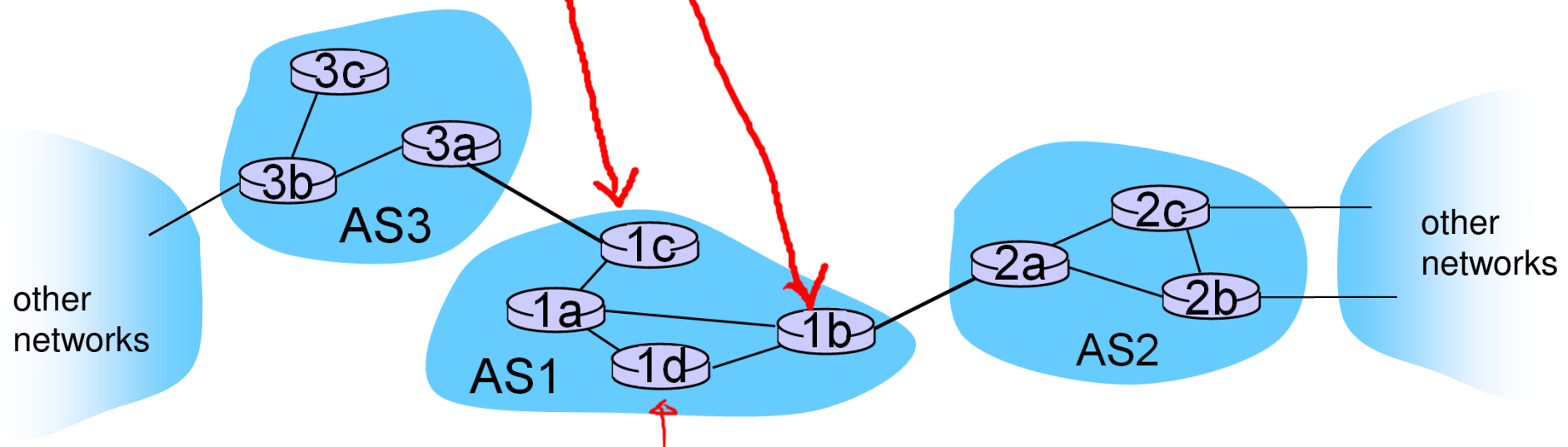
❖ suppose router in AS1 receives datagram destined outside of AS1:

- Q: router^{1d} should forward packet to gateway router, but which one?

AS1 must:

1. learn which dests are reachable through AS2, which through AS3
2. propagate this reachability info to all routers in AS1

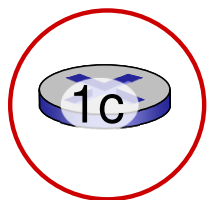
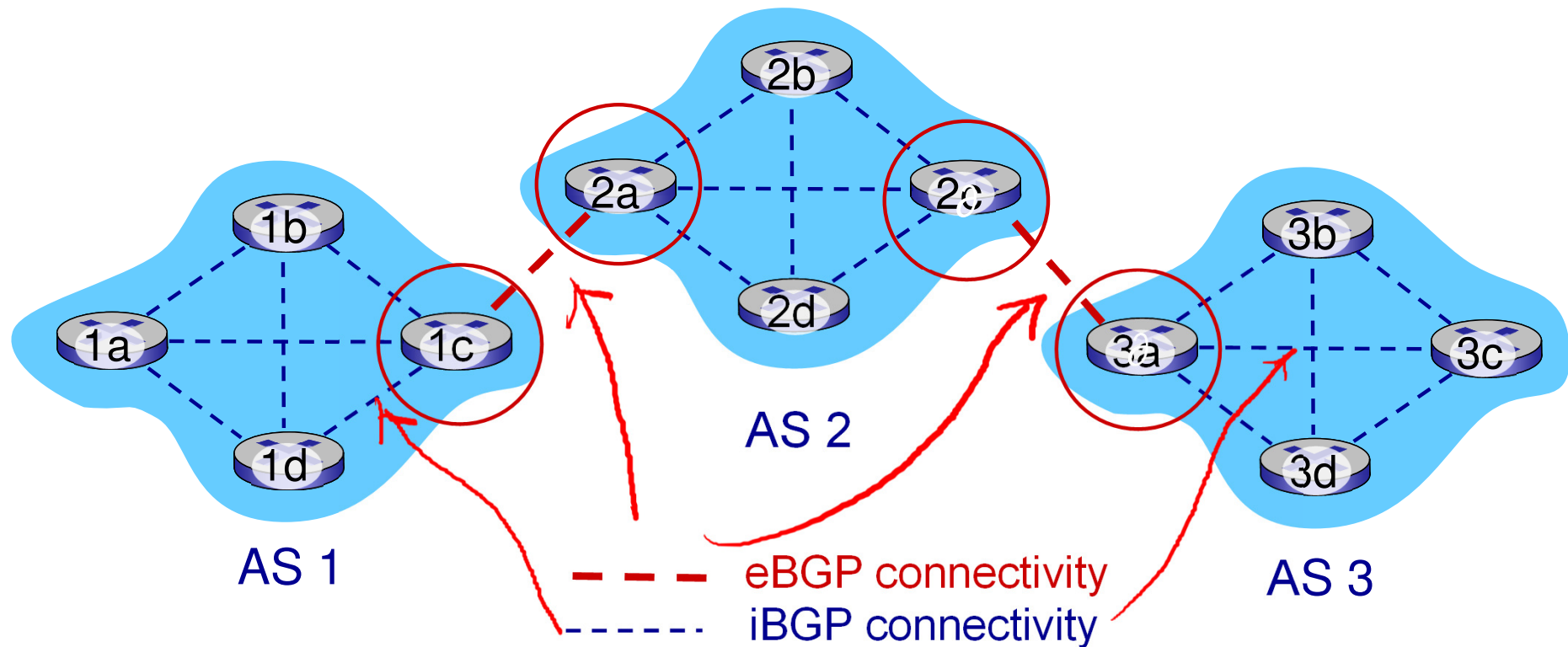
job of inter-AS routing!



Internet inter-AS routing: BGP

- ❖ BGP (Border Gateway Protocol): *the de facto inter-domain* routing protocol
 - “glue that holds the Internet together”
- ❖ BGP provides each AS a means to:
 - External BGP (eBGP): obtain subnet reachability information from neighboring ASs.
 - Internal BGP (iBGP): propagate reachability information to all AS-internal routers.
 - determine “good” routes to other networks based on reachability information and policy.
- ❖ allows subnet to advertise its existence to rest of Internet: “*am here*”

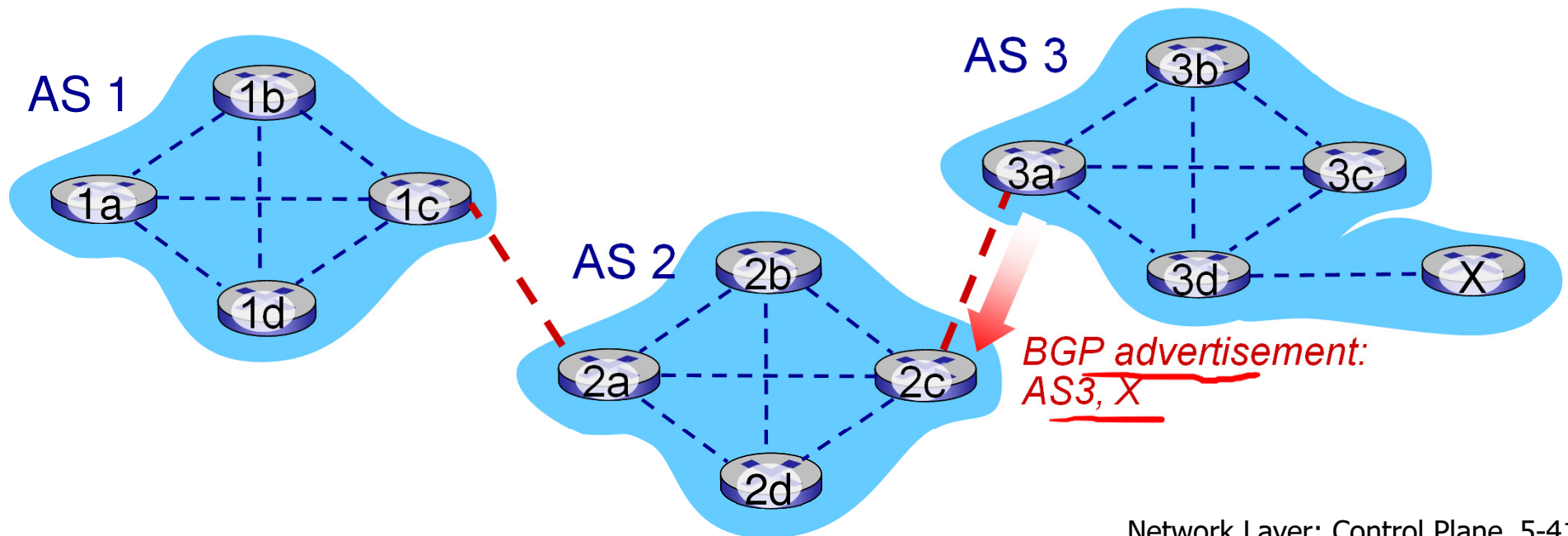
eBGP, iBGP connections



gateway routers run both eBGP and iBGP protocols

BGP basics

- advertisement of paths (e.g. AS3, X)
→ forward promise (e.g. AS2)*
- **BGP session:** two BGP routers (“peers”) exchange BGP messages over semi-permanent TCP connection:
 - advertising *paths* to different destination network prefixes (BGP is a “path vector” protocol)
 - E.g. when AS3 gateway router 3a advertises path AS3, X to AS2 gateway router 2c:
 - AS3 promises to AS2 it will forward datagrams towards X



Path attributes and BGP routes

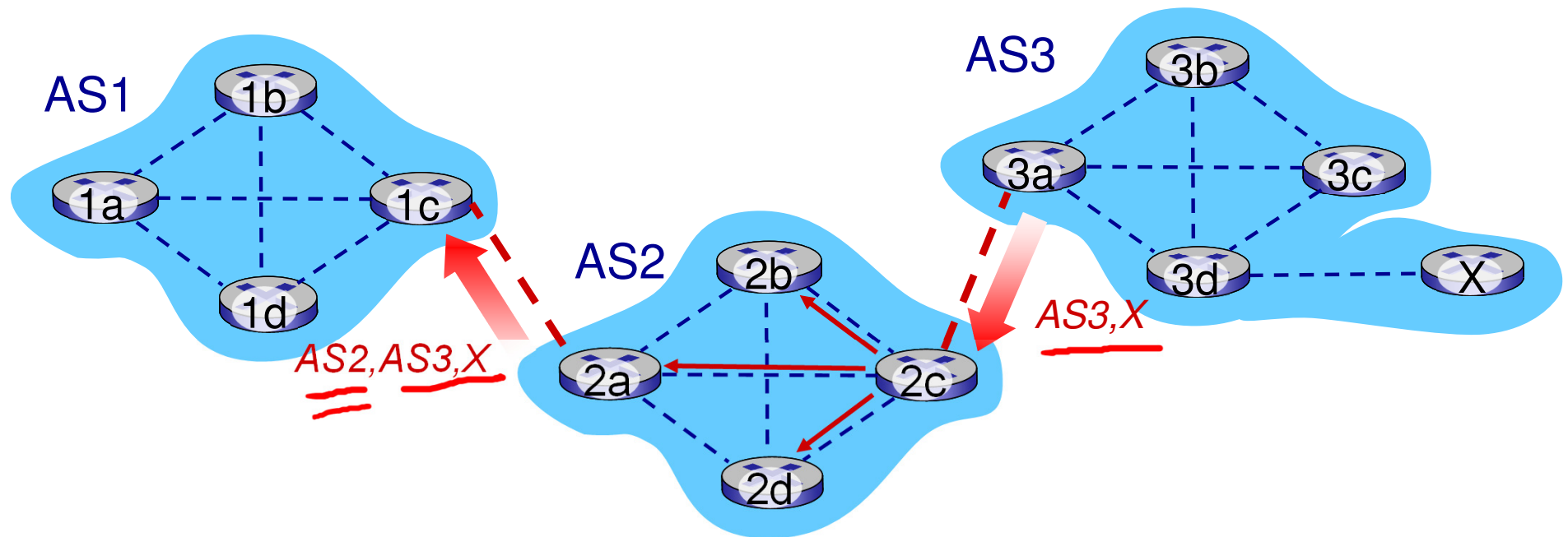
Subject we can reach

- advertised prefix includes BGP attributes
 - *properties*
prefix + attributes = “route”
- two important attributes:
 - AS-PATH: list of ASes through which prefix advertisement has passed
 - NEXT-HOP: IP address of the router interface that begins the AS PATH
- *Policy-based routing*:
 - gateway receiving route advertisement uses *import policy* to accept/decline path (e.g., never route through AS Y).
 - AS policy also determines whether to *advertise* path to other neighboring ASes

BGP messages

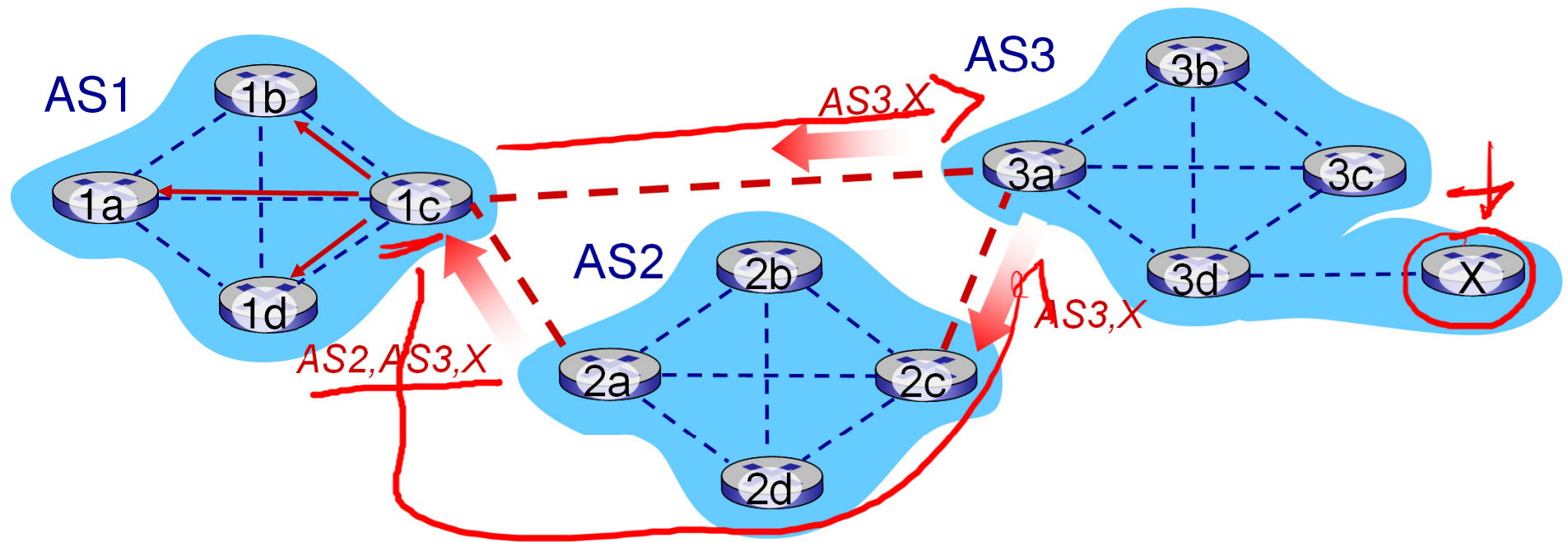
- BGP messages exchanged between peers over TCP connection
- BGP messages:
 - **OPEN:** opens TCP connection to remote BGP peer and authenticates sending BGP peer
 - **UPDATE:** advertises new path (or withdraws old)
 - **KEEPALIVE:** keeps connection alive in absence of UPDATES; also ACKs OPEN request
 - **NOTIFICATION:** reports errors in previous msg; also used to close connection

BGP path advertisement



- AS2 router 2c receives path advertisement AS3,X (via eBGP) from AS3 router 3a
- Based on AS2 policy, AS2 router 2c accepts path AS3,X, propagates (via iBGP) to all AS2 routers
- Based on AS2 policy, AS2 router 2a advertises (via eBGP) path AS2,AS3,X to AS1 router 1c

BGP path advertisement



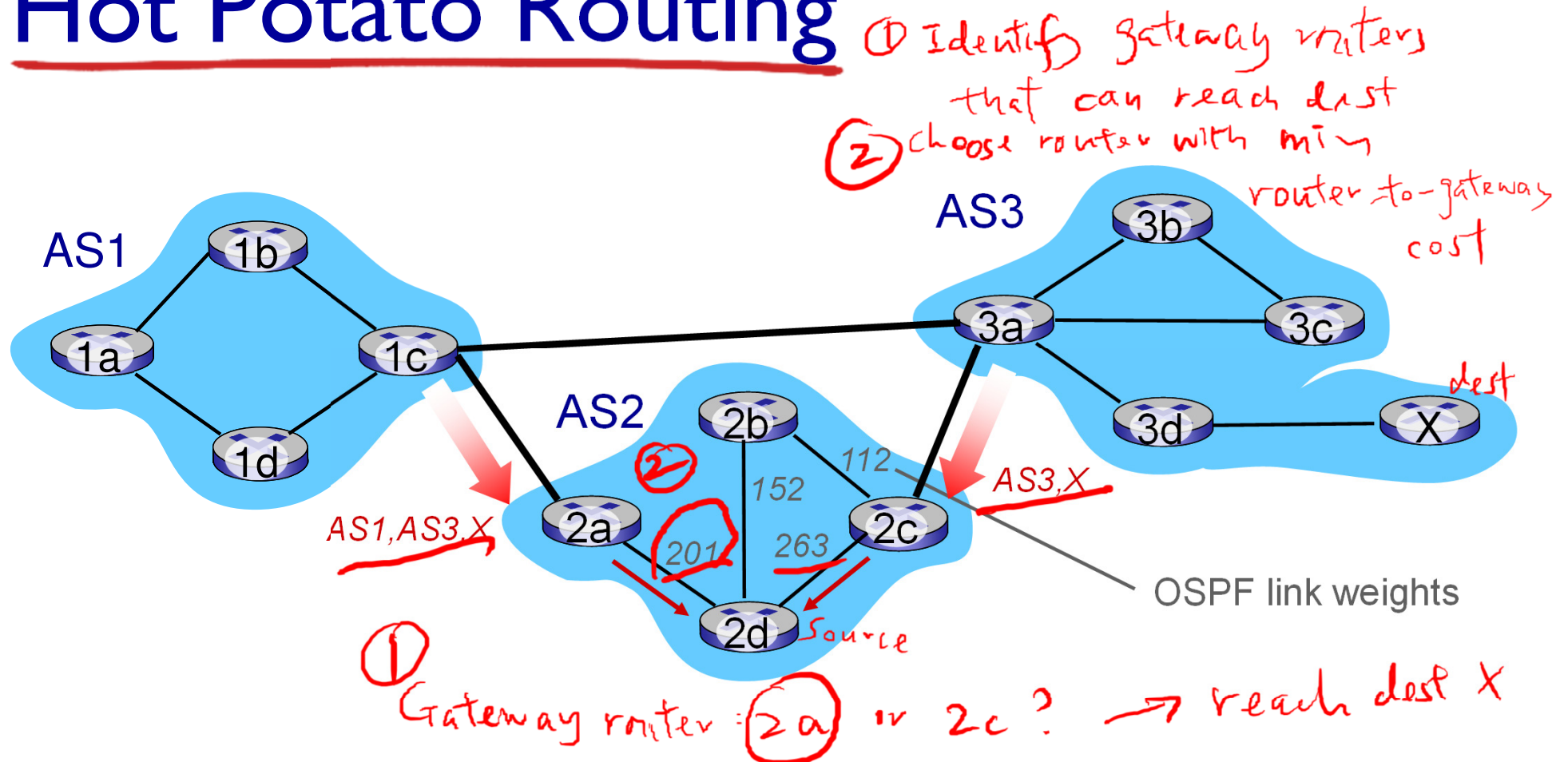
gateway router may learn about multiple paths to destination:

- AS1 gateway router 1c learns path AS2,AS3,X from 2a
- AS1 gateway router 1c learns path AS3,X from 3a
- Which path should a router choose?

BGP route selection

- router may learn about more than one route to destination AS, selects route based on:
 1. local preference value attribute: ^{admin, econ, political} policy decision
 2. shortest AS-PATH AS3, X vs AS2, AS3, X
 3. closest NEXT-HOP router: hot potato routing
 4. additional criteria

Hot Potato Routing



- 2d learns (via iBGP) it can route to X via 2a or 2c
- **hot potato routing**: choose local gateway that has least intra-domain cost (e.g., 2d chooses 2a, even though more AS hops to X): ~~don't worry about inter-domain cost!~~

Why different Intra-, Inter-AS routing?

policy:

- ❖ inter-AS: admin wants control over how its traffic routed, who routes through its net.
- ❖ intra-AS: single admin, so no policy decisions needed

scale:

- ❖ hierarchical routing saves table size, reduced update traffic

performance:

- ❖ inter-AS: policy may dominate over performance
- ❖ intra-AS: can focus on performance

Chapter 5: outline

5.1 introduction

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet: OSPF

5.4 routing among the ISPs: BGP

5.5 The SDN control plane

5.6 ICMP: The Internet
Control Message
Protocol

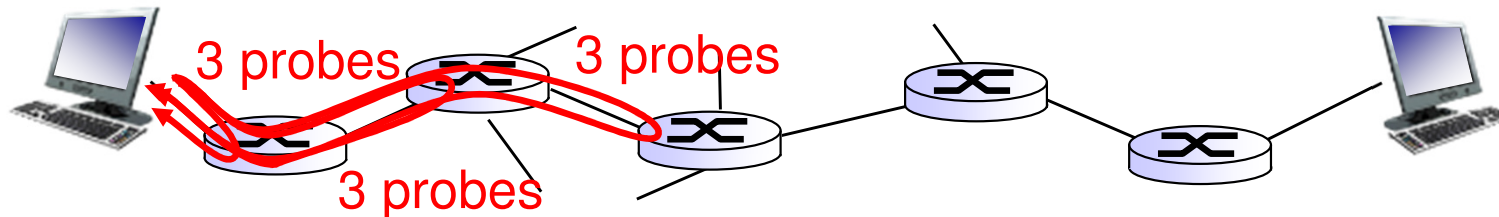
ICMP: internet control message protocol

- ❖ used by hosts & routers to **communicate network-level information**
 - error reporting: unreachable host, network, port, protocol
 - echo request/reply (used by ping)
- ❖ network-layer “above” IP:
 - ICMP msgs carried in IP datagrams
- ❖ **ICMP message**: type, code plus first 8 bytes of IP datagram causing error

Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

Traceroute and ICMP

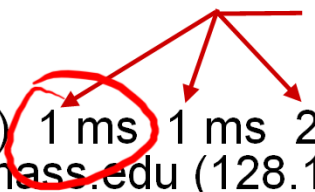
- ❖ `traceroute` program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination
 - router i will return packets to sender
 - sender times interval between transmission and reply.



Traceroute and ICMP

Source *dest*
traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu



1	cs-gw (128.119.240.254)	1 ms	1 ms	2 ms
2	border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)	1 ms	1 ms	2 ms
3	cht-vbns.gw.umass.edu (128.119.3.130)	6 ms	5 ms	5 ms
4	jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)	16 ms	11 ms	13 ms
5	jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)	21 ms	18 ms	18 ms
6	abilene-vbns.abilene.ucaid.edu (198.32.11.9)	22 ms	18 ms	22 ms
7	nycm-wash.abilene.ucaid.edu (198.32.8.46)	22 ms	22 ms	22 ms
8	62.40.103.253 (62.40.103.253)	104 ms	109 ms	106 ms
9	de2-1.de1.de.geant.net (62.40.96.129)	109 ms	102 ms	104 ms
10	de.fr1.fr.geant.net (62.40.96.50)	113 ms	121 ms	114 ms
11	renater-gw.fr1.fr.geant.net (62.40.103.54)	112 ms	114 ms	112 ms
12	nio-n2.cssi.renater.fr (193.51.206.13)	111 ms	114 ms	116 ms
13	nice.cssi.renater.fr (195.220.98.102)	123 ms	125 ms	124 ms
14	r3t2-nice.cssi.renater.fr (195.220.98.110)	126 ms	126 ms	124 ms
15	eurecom-valbonne.r3t2.ft.net (193.48.50.54)	135 ms	128 ms	133 ms
16	194.214.211.25 (194.214.211.25)	126 ms	128 ms	126 ms
17	***			
18	***			
19	fantasia.eurecom.fr (193.55.113.142)	132 ms	128 ms	136 ms

trans-oceanic link

* means no response (probe lost, router not replying)

Q: How to design Traceroute program based on ICMP?

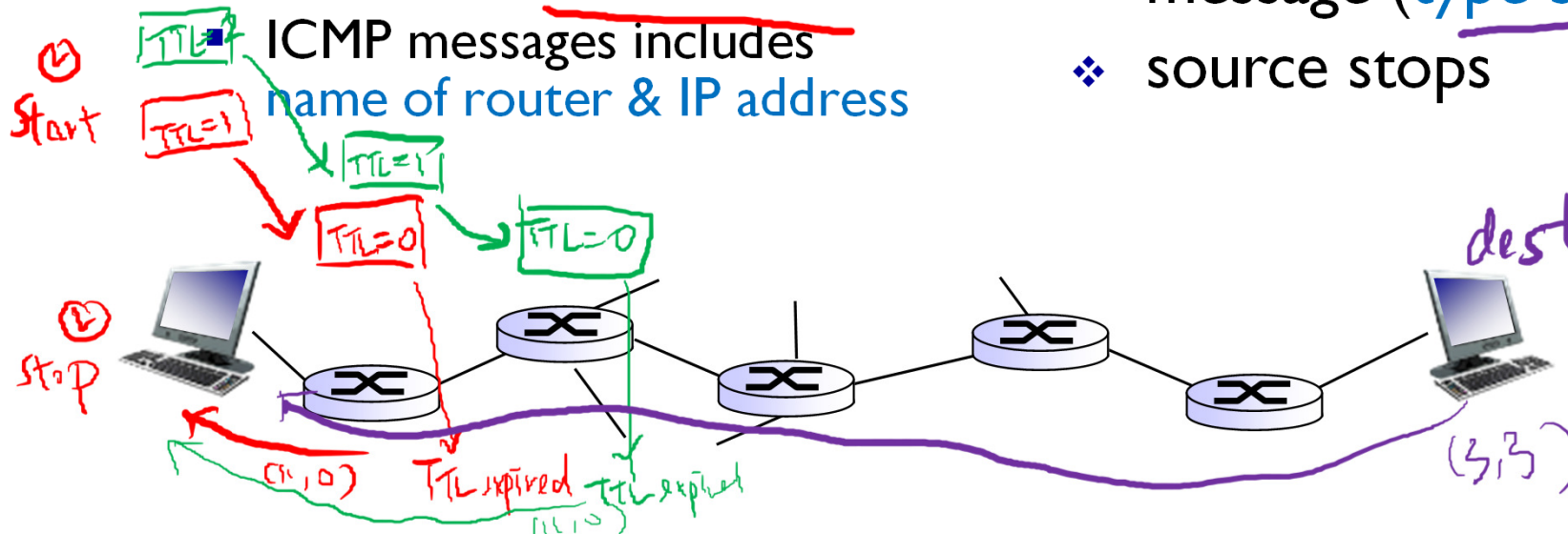
Traceroute and ICMP

- ❖ source sends series of UDP segments to dest
 - first set has TTL = 1
 - second set has TTL=2, etc.
 - unlikely port number
- ❖ when nth set of datagrams arrives to nth router:
 - router discards datagrams
 - and sends source ICMP messages (type 11, code 0)

- ❖ when ICMP messages arrives, source records RTTs

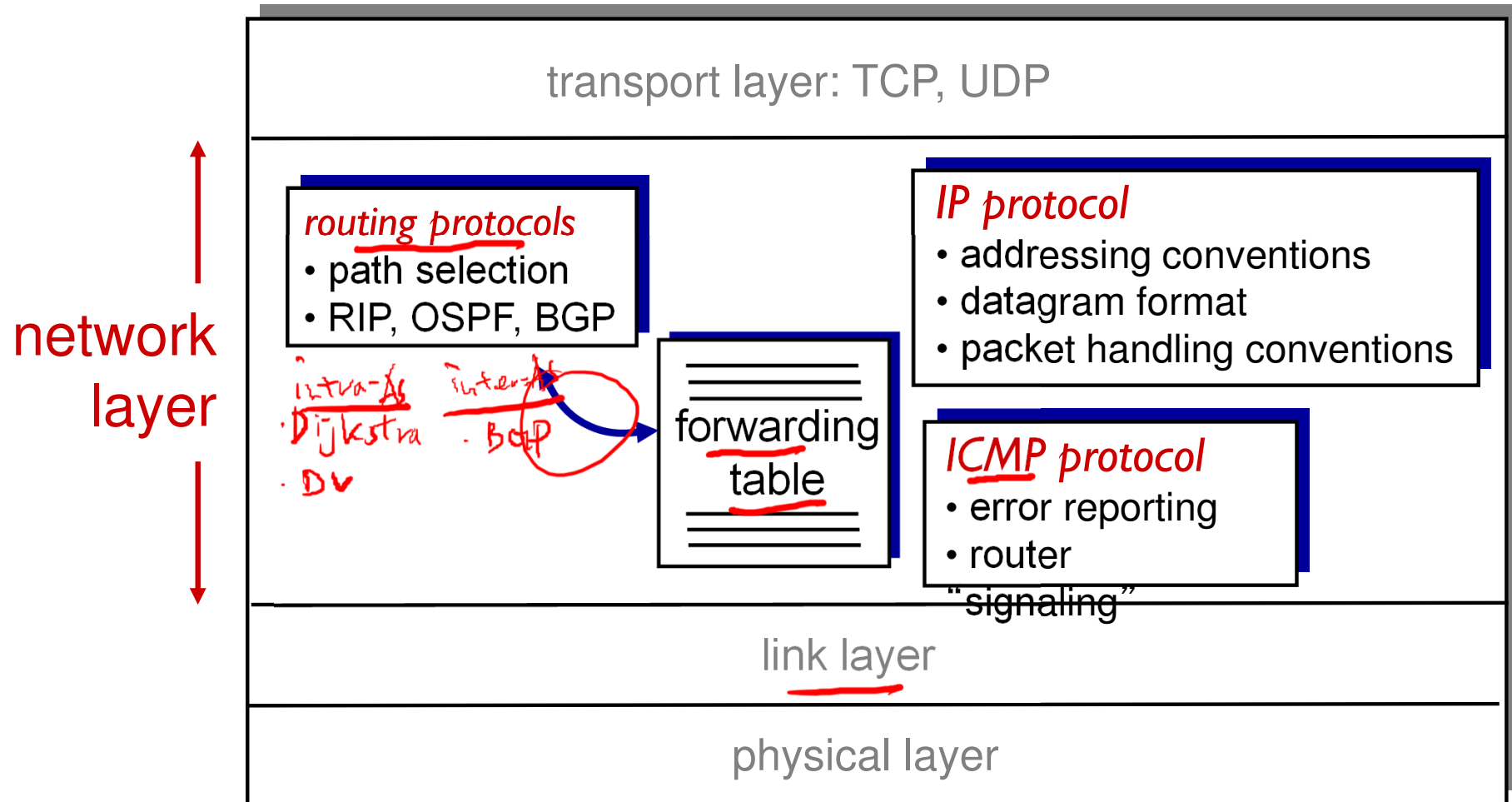
stopping criteria:

- ❖ UDP segment eventually arrives at destination host
- ❖ destination returns ICMP "port unreachable" message (type 3, code 3)
- ❖ source stops



The Internet network layer

host, router network layer functions:



Chapter 5: summary

we've learned a lot!

- approaches to network control plane
 - per-router control (traditional)
 - logically centralized control (software defined networking)
- traditional routing algorithms
 - implementation in Internet: OSPF, BGP
- SDN controllers
 - implementation in practice: ODL, ONOS
- Internet Control Message Protocol

next stop: link layer!